Daryna Dementieva

Postdoctoral Researcher · Responsible AI & Language Technologies

daryna.dementieva@tum.de | dardem.github.io | Google Scholar | LinkedIn | GitHub | Hugging Face
School of Computation, Information and Technology, Technical University of Munich, Garching, Germany

Researcher in Responsible AI and Language Technologies, working on trustworthy evaluation of AI, combating harmful content online, and multilingual, low-resource language technologies — including Ukrainian. Author of 50+ peer-reviewed papers and organiser of the NLP for Positive Impact workshop and the TextDetox and LLMs with Limited Resources shared tasks.

Academic Positions

Postdoctoral Researcher, Technical University of Munich (TUM), *Munich, Germany* 2022 – present
Host: Prof. Dr. Alexander Fraser (since 2024; before: Prof. Dr. Georg Groh). Research: Responsible AI; Reasoning for good and in multilingual context; hate and toxic speech mitigation; explainable AI; multilingual, low-resource, and Ukrainian NLP.

Visiting Researcher, Indian Institute of Technology Kharagpur, *India* February 2023
Group of Prof. Animesh Mukherjee; funded by the Joint India–Germany Research Grant.

Visiting Researcher, Mannheim NLP Group, *Mannheim, Germany* Spring 2022
Group of Prof. Simone Paolo Ponzetto

Research Engineer, Skolkovo Institute of Science and Technology (Skoltech), *Moscow* 2021 – 2022
Led the Text Detoxification project: two ACL papers, one journal paper, and the RUSSE-2022 shared task; results deployed in the MTS.AI chatbot.

Industry Experience

Data Scientist, Visiology, *Moscow* 2017 – 2019
NLP2SQL: chatbot assistant for Business Intelligence. Presented at the Business Intelligence 2.0 conference (1.5k attendees) as the first Russian BI chatbot and deployed as ViTalk.

Data Science Intern, Beiersdorf, *Hamburg, Germany* 2018
Concept design and prototype of an automatic trend extractor for social-media texts.

Education

Ph.D. in Computer Science, Skoltech, *Moscow* 2019 – 2022
Thesis: *Methods for Fighting Harmful Multilingual Information*. Supervisor: Asst. Prof. Alexander Panchenko. Defence: information and video.

M.Sc. in Applied Mathematics and Physics, Moscow Institute of Physics and Technology (MIPT), *Moscow* 2017 – 2019
Thesis: *News Aggregation and Personalization for Professional Groups*. Supervisor: Prof. Konstantin Vorontsov.

B.Sc. in Applied Mathematics, Taras Shevchenko National University of Kyiv, *Kyiv, Ukraine* 2013 – 2017

Grants, Fellowships & Awards

Grants & Fellowships

2024 **Friedrich Schiedel Fellowship for Technology in Society**, TUM Think Tank: € 20,000 to explore *Harmful Speech Proactive Moderation* (video pitch).

2024 **Abbe Grant**, Carl-Zeiss-Stiftung: fully funded participation in the 11th Heidelberg Laureate Forum (top 1% of applicants).

2024 **Toloka.ai Research Grant**: $ 3,000 for multilingual and cross-lingual annotation projects.

2023 **Joint India–Germany Research Grant**: *Deep neural multilingual models to combat online hate content*; funded research stay at IIT Kharagpur.

Awards & Honours

2025 **Best Resource Paper Award**, ACL 2025, for *BRIGHTER: Bridging the Gap in Human-Annotated Textual Emotion Recognition Datasets for 28 Languages*.

2025 **Invited keynote**, The Future of NLP symposium, HITS Heidelberg, on the recommendation of Prof. Michael Strube.

2025 **Featured** in the Women in AI Scientist Spotlight series for research on combating fake news and mitigating hate speech.

2023 **Winning system**, SemEval-2023 Task 4 (Human Value Detection), with M.Sc. student Daniel Schroter.

2023 **Exceptional Reviewer**, ACL Rolling Review (October 2023 cycle).

2022 **Skoltech Highest Scholarship** (2022–2023) for outstanding research achievements (top 1%).

Competitions

2021 **Junction 2021**, Empathy Challenge: third prize — text detoxification for a gaming use case.

2019 **Junction 2019**, Ericsson Challenge: second prize — optimal client–taxi matching.

2018 **P&G IT Business Challenge**: winner, representing the Eastern Europe region in Athens, Greece.

2015 **Innopolis Programming Contest**: winner; educational grant (top 1%).

Publications

Also on Google Scholar. * equal contribution.

2026

[1] Antonia Karamolegkou, Angana Borah, ..., Anders Søgaard, Alexander Fraser, Zhijing Jin, Rada Mihalcea, Joel Tetreault, **Daryna Dementieva**. "NLP for Social Good: A Survey and Outlook of Challenges, Opportunities and Responsible Deployment." *EACL 2026*.

[2] Francielle Vargas, Jackson Trager, Diego Alves, Surendrabikram Thapa, Matteo Guida, Berk Atil, **Daryna Dementieva**, Andrew Smart, Ameeta Agrawal. "Self-Explaining Hate Speech Detection with Moral Rationales." *arXiv preprint, 2026*.

[3] Ekaterina Artemova, Laurie Burchell, **Daryna Dementieva**, Shu Okabe, Mariya Shmatova, Pedro Ortiz Suarez. "Low-Resource, High-Impact: Building Corpora for Inclusive Language Technologies." *LREC 2026, Tutorial*.

2025

[4] **Daryna Dementieva**, Evgeniya Sukhodolskaya, Alexander Fraser. "CrossNews-UA: A Cross-lingual News Semantic Similarity Benchmark for Ukrainian, Polish, Russian, and English." *arXiv preprint, 2025*.

[5] Shu Okabe, **Daryna Dementieva**, Marion Weller-Di Marco, Lukas Edman, Katharina Haemmerl, Marko Měškank, Anita Hendrichowa, Alexander Fraser. "Findings of the WMT 2025 Shared Task LLMs with Limited Resources for Slavic Languages: MT and QA." *WMT 2025 @ EMNLP 2025, Shared Task Findings*.

[6] **Daryna Dementieva**, Nikolay Babakov, Alexander Fraser. "EmoBench-UA: A Benchmark Dataset for Emotion Detection in Ukrainian." *Findings of EMNLP 2025*.

[7] Faeze Ghorbanpour, **Daryna Dementieva**, Alexander Fraser. "Data-Efficient Hate Speech Detection via Cross-Lingual Nearest Neighbor Retrieval with Limited Labeled Data." *EMNLP 2025*.

[8] Mariia Bogatyreva, Pascal Gaertner, Quim Ribas, **Daryna Dementieva**, Alexander Fraser. "QiMP at SemEval-2025 Task 11: Optimizing Text-based Emotion Classification in English Beyond Traditional Methods." *SemEval 2025 @ ACL 2025*.

[9] Viola De Ruvo, Arianna Muti, **Daryna Dementieva**, Debora Nozza. "DETOXIFY-IT: An Italian Parallel Dataset for Text Detoxification." *WOAH 2025 @ ACL 2025*.

[10] Faeze Ghorbanpour, **Daryna Dementieva**, Alexander Fraser. "Can Prompting LLMs Unlock Hate Speech Detection across Languages? A Zero-shot and Few-shot Study." *WOAH 2025 @ ACL 2025*.

[11] Naquee Rizwan, Seid Muhie Yimam, **Daryna Dementieva**, et al. "HatePRISM: Policies, Platforms, and Research Integration. Advancing NLP for Hate Speech Proactive Mitigation." *Findings of ACL 2025*.

[12] Shamsuddeen Hassan Muhammad, Nedjma Ousidhoum, ..., **Daryna Dementieva**, ..., Saif M. Mohammad. "BRIGHTER: BRIdging the Gap in Human-Annotated Textual Emotion Recognition Datasets for 28 Languages." *ACL 2025*. **Best Resource Paper Award.**

[13] **Daryna Dementieva**, Nikolay Babakov, Amit Ronen, Abinew Ali Ayele, Naquee Rizwan, Florian Schneider, Xintong Wang, Seid Muhie Yimam, Daniil Moskovskiy, Elisei Stakovskii, Eran Kaufman, Ashraf Elnagar, Animesh Mukherjee, Alexander Panchenko. "Multilingual and Explainable Text Detoxification with Parallel Corpora." *COLING 2025*.

[14] **Daryna Dementieva**, Valeriia Khylenko, Georg Groh. "Cross-lingual Text Classification Transfer: The Case of Ukrainian." *COLING 2025*.

[15] **Daryna Dementieva**, Vitaly Protasov, Nikolay Babakov, Naquee Rizwan, Ilseyar Alimova, Caroline Brun, Vasily Konovalov, Arianna Muti, Chaya Liebeskind, Marina Litvak, Debora Nozza, Shehryaar Shah, Sotaro Takeshita, Natalia Vanetik, Abinew Ali, Florian Schneider, Xintong Wang, Seid Muhie, Ashraf Elnagar, Animesh Mukherjee, Alexander Panchenko. "Overview of the Multilingual Text Detoxification Task at PAN 2025." *CLEF 2025 Working Notes (PAN)*.

2024

[16] **Daryna Dementieva**, Daniil Moskovskiy, Nikolay Babakov, et al. "Overview of the Multilingual Text Detoxification Task at PAN 2024." *CLEF 2024 Working Notes (PAN)*.

[17] Cem Üyük, Danica Rovó, Shaghayeghkolli Shaghayeghkolli, Rabia Varol, Georg Groh, **Daryna Dementieva**. "Crafting Tomorrow's Headlines: Neural News Generation and Detection in English, Turkish, Hungarian, and Persian." *NLP4PI 2024 @ EMNLP 2024*.

[18] **Daryna Dementieva**, Nikolay Babakov, Alexander Panchenko. "MultiParaDetox: Extending Text Detoxification with Parallel Data to New Languages." *NAACL 2024*.

[19] **Daryna Dementieva**, Valeriia Khylenko, Nikolay Babakov, Georg Groh. "Toxicity Classification in Ukrainian." *WOAH 2024 @ NAACL 2024*.

2023

[20] Enkelejda Kasneci, Kathrin Seßler, Stefan Küchemann, Maria Bannert, **Daryna Dementieva**, ..., Gjergji Kasneci. "ChatGPT for Good? On Opportunities and Challenges of Large Language Models for Education." *Learning and Individual Differences (journal), 2023*.

[21] Daniel Schroter, **Daryna Dementieva**, Georg Groh. "Adam-Smith at SemEval-2023 Task 4: Discovering Human Values in Arguments with Ensembles of Transformer-based Models." *SemEval 2023 @ ACL 2023*. **Winning system.**

[22] Adam Rydelek, **Daryna Dementieva**, Georg Groh. "AdamR at SemEval-2023 Task 10: Solving the Class Imbalance Problem in Sexism Detection with Ensemble Learning." *SemEval 2023 @ ACL 2023*.

[23] **Daryna Dementieva**, Mikhail Kuimov, Alexander Panchenko. "Multiverse: Multilingual Evidence for Fake News Detection." *Journal of Imaging, 2023*.

[24] **Daryna Dementieva**, Nikolay Babakov, Alexander Panchenko. "Detecting Text Formality: A Study of Text Classification Approaches." *RANLP 2023*.

[25] **Daryna Dementieva**, Daniil Moskovskiy, David Dale, Alexander Panchenko. "Exploring Methods for Cross-lingual Text Style Transfer: The Case of Text Detoxification." *IJCNLP-AACL 2023*.

[26] Edoardo Mosca, **Daryna Dementieva**, Tohid Ebrahim Ajdari, Maximilian Kummeth, Kirill Gringauz, Yutong Zhou, Georg Groh. "IFAN: An Explainability-Focused Interaction Framework for Humans and NLP Models." *IJCNLP-AACL 2023, System Demonstrations*.

2022

[27] Varvara Logacheva*, **Daryna Dementieva***, Sergey Ustyantsev, Daniil Moskovskiy, David Dale, Irina Krotova, Nikita Semenov, Alexander Panchenko. "ParaDetox: Detoxification with Parallel Data." *ACL 2022*.

[28] Daniil Moskovskiy, **Daryna Dementieva**, Alexander Panchenko. "Exploring Cross-lingual Text Detoxification with Large Multilingual Language Models." *ACL 2022, Student Research Workshop*.

[29] **Daryna Dementieva***, Varvara Logacheva*, Irina Nikishina, Alena Fenogenova, David Dale, Irina Krotova, Nikita Semenov, Tatiana Shavrina, Alexander Panchenko. "RUSSE-2022: Findings of the First Russian Detoxification Shared Task Based on Parallel Corpora." *Dialogue 2022*.

[30] Varvara Logacheva, **Daryna Dementieva**, ..., Alexander Panchenko. "A Study on Manual and Automatic Evaluation for Text Style Transfer: The Case of Detoxification." *HumEval 2022 @ ACL 2022*.

2021

[31] David Dale, Anton Voronov, **Daryna Dementieva**, Varvara Logacheva, Olga Kozlova, Nikita Semenov, Alexander Panchenko. "Text Detoxification using Large Pre-trained Neural Models." *EMNLP 2021*.

[32] **Daryna Dementieva**, Alexander Panchenko. "Cross-lingual Evidence Improves Monolingual Fake News Detection." *ACL-IJCNLP 2021, Student Research Workshop*.

[33] **Daryna Dementieva**, Daniil Moskovskiy, Varvara Logacheva, David Dale, Olga Kozlova, Nikita Semenov, Alexander Panchenko. "Methods for Detoxification of Texts for the Russian Language." *Multimodal Technologies and Interaction (journal), 2021*.

Teaching

since 2026	**Responsible AI and Language Technologies**, seminar, TUM — organiser.
since 2024	**Lab Courses in NLP and Large Language Models** (B.Sc. & M.Sc., ~50 students per semester), TUM — organiser.
2022 – 2024	**Natural Language Processing**, lecture (~200 students), TUM — lecturer.
2022 – 2024	**M.Sc. Lab Course: XAI for Machine Learning**, TUM — project supervision.
2019 – 2021	**Statistical NLP** and **Neural NLP**, Skoltech — seminars and project supervision.
2026	**Tutorial**: Low-Resource, High-Impact: Building Corpora for Inclusive Language Technologies, LREC 2026 — co-presenter.

Student Supervision

Ph.D.

Faeze Ghorbanpour (Ph.D., TUM, for 2024). *Efficient Methods for Hate Speech Mitigation*. Papers: EMNLP 2025, WOAH 2025

M.Sc. and B.Sc. theses

[Name] ([M.Sc./B.Sc.], TUM, [2025]). *[Thesis title — add students from 2025–2026]*.

Evgeniya Sukhodolskaya (M.Sc., TUM, 2024). *A Unified Approach to Cross-lingual News Similarity Evaluation: from Crowdsourcing to Large Language Models Optimization*. Paper: CrossNews-UA, 2025

Diego Miguel Lozano (M.Sc., TUM, 2024). *Explainable Textual Similarity Metrics via Equivalent and Erroneous Span Detection*.

Arda Andirin (M.Sc., TUM, 2024). *Explainable Fact-Checking in News Datasets with Large Language Models*.

Shaghayegh Koli (M.Sc., TUM, 2024). *Fake News and Machine-generated Text Detection for Persian and English*. Paper: NLP4PI 2024

Valeriia Khylenko (M.Sc., TUM, 2024). *Cross-lingual Text Classification Knowledge Transfer to the Ukrainian Language*. Papers: COLING 2025, WOAH 2024

Adam Rydelek (M.Sc., TUM, 2023). *Sexism in Social Media — Explainable Detection Using Deep Learning Models*. Paper: SemEval 2023

Daniel Schroter (M.Sc., TUM, 2023). *Discovering Human Values Behind Arguments — A Case for Transformer-based Models and Explainable AI*. Paper: SemEval 2023, **winning system**

Yen-Yu Chang (M.Sc., TUM, 2023). *Exploration of Approaches to Counter Hate Speech: The Case of Sexist Speech*.

Yuliana Poliakova (B.Sc., TUM, 2023). *Detecting the Persuasion Techniques in Online News*.

Daniil Moskovskiy (M.Sc., Skoltech, 2022). *Supervised Text Detoxification Methods*. Paper: ACL 2022 SRW

Mikhail Kuimov (M.Sc., Skoltech, 2022). *Multilingual and Cross-lingual Approaches to Processing of News Texts*. Paper: SemEval 2022

Talks & Presentations

Invited talks, keynotes, and panels

2025	*Multilingual and Responsible Language Technologies for Social Good*. The Future of NLP, HITS, Heidelberg, Germany (keynote)
Jun 2025	*Methods for Multilingual Harmful Speech Mitigation*. EALM Workshop: Ethics and Alignment of (Large) Language Models, Marseille, France (keynote)
Apr 2025	*Modern NLP for Proactive Harmful Content Moderation*. PyCon DE & PyData, Darmstadt, Germany (talk)
Mar 2025	*Methods for Multilingual Harmful Speech Mitigation*. The Alan Turing Institute, Seminar, London, UK (invited talk)
Apr 2024	*LLMs from a Socio-technical System Perspective*. alignAI, Munich, Germany (panel)
Apr 2024	*How to Do Monolingual, Multilingual, and Cross-lingual Text Classification*. PyCon DE & PyData, Berlin, Germany (talk)
Jun 2023	*Methods for Fighting Harmful Multilingual Information*. NLP Group, University of Mannheim, Germany (invited talk)
Apr 2023	*Methods for Text Style Transfer: Text Detoxification Case*. PyCon DE & PyData, Berlin, Germany (talk)
Nov 2022	*Methods for Fighting Harmful Multilingual Information*. Language Technology Group, University of Hamburg, Germany (invited talk)
Oct 2022	*NLP for Social Good: Methods for Fighting Harmful Multilingual Textual Content*. MunichNLP Meetup, Munich, Germany (talk)
Oct 2022	*Methods for Text Style Transfer: Text Detoxification Case*. MilaNLP, Bocconi University (online) (invited talk)
Jun 2022	*Methods for Fighting Harmful Multilingual Information*. CIS, LMU Munich, Germany (invited talk)
Jul 2021	*Fake News Detection using Multilingual Evidence*. Trustworthy AI, Moscow (talk)
May 2019	*Virtual Assistant ViTalk or Human Business Intelligence*. DataFest, Moscow (talk)

Academic Service

Workshops, shared tasks, and tutorials

2026	Lead organiser, 5th Workshop on NLP for Positive Impact, EMNLP 2026.
2026	Co-organiser, tutorial Low-Resource, High-Impact: Building Corpora for Inclusive Language Technologies, LREC 2026.
2025 – 2026	Co-organiser, shared task LLMs with Limited Resources for Slavic Languages, WMT @ EMNLP.
2025	Lead organiser, 4th Workshop on NLP for Positive Impact, ACL 2025.
2025	Co-organiser, SemEval-2025 Task 11: Bridging the Gap in Text-Based Emotion Detection (28

	languages).
2024 – 2025	Lead organiser, Multilingual Text Detoxification (TextDetox) shared task, PAN @ CLEF 2024 and 2025.
2024	Co-organiser, 3rd Workshop on NLP for Positive Impact, EMNLP 2024.
2022	Lead organiser, RUSSE-2022: the first Russian text detoxification shared task, Dialogue 2022.

Area chair and reviewing

since 2024	**Area Chair**, ACL Rolling Review.
ARR	February 2024; October 2023 (Exceptional Reviewer); August 2023; June 2023.
ACL	2020, 2021, 2022, 2023.
EMNLP	2020, 2021, 2022, 2023, 2024.
WOAH	2023, 2024.
HCI	2023, 2024.
VLDB	2021, 2023.
Other	LREC 2024, ECAI 2024, ESWA 2024, DSAA 2020.

Open Source

My open-sourced models and datasets have reached **over 3M downloads** altogether.

- **Founder** of textdetox (multilingual text detoxification, 15 languages) and ukr-detect (Ukrainian text classification).
- **Founder** of tum-nlp: state-of-the-art human values detector (SemEval-2023 winner), target-aware counter-speech generator, sexism detector.
- **Co-admin** of s-nlp: the first English and Russian ParaDetox datasets; English and Russian toxicity classifiers; state-of-the-art detoxification models for English and Russian; multilingual formality classifier.
- Overview of the text detoxification research line: github.com/dardem/text_detoxification.

Outreach & Community

Organiser of the Munich NLP community and meetups. Member of Women in AI (Munich and Germany chapters); mentor at Women in NLP Munich and CyberMentor, supporting young women in STEM. Regular speaker at PyCon DE & PyData.

Summer & Winter Schools

Sep 2023	ELLIS Summer School on Large-Scale AI for Research and Industry, Modena, Italy.
Jan 2022	Advanced Language Processing Winter School (ALPS), Grenoble, France.
Jan 2021	Advanced Language Processing Winter School (ALPS), online.
Jun 2020	Machine Learning Summer School (MLSS), Tübingen, online.

Languages & Interests

Ukrainian and **Russian** (native), **English** (professional), **German** (intermediate).

Reading, running, pilates and yoga, swimming, windsurfing, hiking, politics and feminism, travelling (32 countries), photography.